

# Anshul Nagar

## Curriculum Vitae

Daejeon South Korea, 35275

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### Education

- 2022–present **Ph.D. student in Electrical Engineering**, *Chungnam National University*
- 2016–2021 **Integrated B.S.-M.S, majored in Chemistry**, *Indian Institute of Science Education and Research, Mohali*, CPI: 7.7/10
- 2014-15 **Class 12<sup>th</sup> (Senior Secondary Examination, RBSE)**, *Tilak Public School, Kota (Rajasthan)*, Aggregate: 89.4%
- 2012-13 **Class 10<sup>th</sup> (Secondary Examination, RBSE)**, *Decent Public School, Baran (Rajasthan)*, Aggregate: 85.4%

### Publications/Preprints

- **Impedance-based health indicators determination for degradation diagnosis and fault identification of lithium-ion batteries under extreme operating conditions**  
Authors: Pyeong-Yeon Lee, **Anshul Nagar**, Kisoo yoo, Prof Johoong Kim
- **ReaxFF and UFF Molecular Dynamic Simulation of SEI components in lithium-ion batteries**  
Authors: **Anshul Nagar**, Dr.Akhil Garg, Surinder Singh, Prof Johoong Kim

### Projects

- July 2024 – **Physics informed Neural Networks**, *Chungnam National University, Daejeon*
- Present ○ Worked on integrating physics-based battery degradation models with neural networks using Physics-Informed Neural Networks (PINNs) for accurate capacity fade prediction.
- Sept 2022 – **Battery Management System Development**, *Chungnam National University, Daejeon*
- June 2024 ○ Collaborative project with Hyundai integrating physics-based models and ML to predict SoC/SoH and critical battery states.
- May 2021 – **Molecular Dynamics of SEI Components**, *Indian Institute of Technology (IIT) Delhi*
- Dec 2021 ○ ReaxFF/UFF simulations of Li-ion diffusion through inorganic SEI at varied temperatures/densities; explored alkali doping. Used AMS (AMSjobs) and DFT workflows.
- Aug 2020 – **Master's Thesis: Conical Intersections in Triatomic Systems**, *Indian Institute of Science Education and Research*, Supervisor: Dr. K. R. Shamasundar
- Apr 2021 ○ Computed energy eigen-matrices to analyze accidental degeneracies vs. group-theory predictions; used perturbation theory and symmetry methods.

May – July 2019 **Computational Chemistry Project**, *Indian Institute of Science Education and Research*

- Geometry optimizations in Gaussian; studied Hartree–Fock, DFT, and *ab initio* methods.

May – July 2018 **Perovskite Solar Cell Synthesis**, *Indian Institute of Science Education and Research*, Supervisor: Dr. Arijit Kumar De

- Synthesized/characterized perovskite cells; surveyed approaches to improve photovoltaic efficiency.

## Teaching Experience

- **Teaching Assistant** for the course CHM617: Chemical Dynamics and Non-adiabatic Interactions.

## Fellowship

- **Received DST-INSPIRE-SHE scholarship** for the period 2016-2021.

## Online Courses Completed/Enrolled

- **Massachusetts Institute of Technology MIT-Open-Course-Ware course**, "[Electrochemical Energy Systems](#)"  
Note: Certification was not provided.
- **University of Colorado Boulder, COURSERA course**, "[Battery Management System](#)" taught by 'Gregory Plett'
- **École Polytechnique, COURSERA course**, "[Density Functional Theory](#)" taught by Francesco Sottile and Lucia Reining

## Computer Skills

- Python, Matlab, LaTeX, Excel/Google Sheets, JSON/CSV handling, Jupyter, Gaussian, AMSjobs, React.js.

## Language Skills

- English (Fluent)
- Hindi (Native)
- Marvadi (Native)
- Korean (fluent – able to understand task instructions, labeling guidelines, and daily communication)

## Workshops

2021 **Symposium on Atomistic Modelling and Simulation**, Organised by Nyro Research Group, Online, 7<sup>th</sup>-11<sup>th</sup> June